

Course: B. Tech.

Branch: Chemical

Semester: III

Subject Code & Name: Process Calculations BTCHC303

Max Marks: 60

Date:

Duration: 3 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed. Assume suitable data wherever necessary and mention it clearly.

		(Level/ CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Solve the following.		6
a.	What are the similarities and differences between chemist and chemical engineers?	U	
b.	i) Convert 10 m/min to km/h. ii) Convert volumetric flow rate 3 m ³ /s to l/s.	R, U, A	
c.	Calculate the kilogram of Na of which the amount is specified as 3 katom.	U, A	
B)	Solve the following.		6
a.	Convert 88 kg of carbon dioxide into its amount in molar units.	U, A	
b.	How many moles of H ₂ SO ₄ will contain 64 kg S?	U, A	
c.	Find grams of HCl needed to prepare 1 liter 2 N HCl solution.	U, A	
C)	Solve the following.		6
a.	98 grams of sulphuric acid (H ₂ SO ₄) are dissolved in water to prepare one liter of solution. Find normality and molarity of solution.	U, A	
b.	An aqueous solution of sodium chloride is prepared by dissolving 25 kg of sodium chloride in 100 kg of water. Determine weight % composition of solution.	U, A	
Q.2	Solve Any Two of the following.		12
A)	Formaldehyde is produced from methanol in a catalytic reactor. The production rate of formaldehyde is 1000 kg/h. If the conversion of methanol is 65%. Calculate the required feed rate of methanol.	U, A	6
B)	The spent acid from a nitrating process contains 21 % HNO ₃ , 55 % H ₂ SO ₄ and 24% H ₂ O by weight. This acid is to be concentrated to contain 28% HNO ₃ and 62% H ₂ SO ₄ by addition of concentrated sulphuric acid containing 93% H ₂ SO ₄ and concentrated nitric acid containing 90 % HNO ₃ . Calculate the weights of spent acid, concentrated sulphuric acid and concentrated nitric acid that must be combined to obtain 100 kg of the desired mixture.	U, A	6
C)	A combustion reactor is fed with 50 kmol/h of butane and 2000 kmol/h of air. Calculate the % excess air used and composition of the gases leaving combustion reactor assuming complete combustion of butane.	U, A	6
Q. 3	Solve Any Two of the following.		12

- A) A mixture H_2 and O_2 contains 11.1 % H_2 by weight. Calculate average molecular weight of gas mixture and partial pressure of O_2 and H_2 at 100 kPa and 303 K (30°C). U, A 6
- B) 1 kmol of a mixture containing 0.4 kmol of N_2 and 0.6 kmol of C_2H_6 occupies a volume of 90 liters at 50°C. What is the pressure in the vessel? Use Van Der Waal equation. Data is given below. U, A 6

Components	Tc, K	Pc, Mpa
C_2H_6	305.42	4.880
N_2	126.09	3.394

- C) Define humidification and Dehumidification. Find the absolute humidity, % RH and dew point using the psychrometric chart using given data : The DB and WB temperatures on a particular day in satara are observed to be 308 K (35°C) and 299 K (26°C) respectively. R, U, A 6

Q.4 Solve Any Two of the following. 12

- A) Solve the following 6

- a. Temperature of oxygen is raised from 350 K to 1500 K. calculate the amount of heat that must be supplied for raising the temperature of 1 kmol oxygen using data given below. $= a + bT + CT^2 + dT^3$ U, A

Gas	a	b x 10 ⁻³	c x 10 ⁻⁶	d x 10 ⁻⁹
O_2	26.0257	11.7551	-2.3426	-0.5623

- b. The gas having the following composition by volume is at temperature of 733 K. $SO_2 = 7.09\%$, $O_2 = 10.55\%$, $SO_3 = 0.45\%$ and $N_2 = 81.91\%$. calculate the heat content of 1 kmol gas mixture over 298 K using the mean molal heat capacity data given below: U, A

Gas	between 773 - 298 K, KJ /(kmol.K)
SO_2	46.8921
O_2	31.4554
SO_3	66.2812
N_2	29.9272

- B) Solve the following 6

- a. Calculate the change in enthalpy between reactants and products if both are at 298 K and if 5 mol of ethylene oxide is produced as per the following reaction: U, A
 $C_2H_4(g) + 0.5 O_2(g) \rightarrow C_2H_4O(g)$

Component	Δ kJ/mol at 298.15 K
$C_2H_4(g)$	52.50
$C_2H_4O(g)$	-52.63

- b. Chlorinated diphenyl is heated from 313 K (40 °C) to 533 K (280 °C) in an indirectly fired heater at the rate of 4000 kg/h. Calculate the heat require to be added to the fluid in the heater. The heat capacity of the fluid in this temperature U, A

range is given below.

$C = 0.7511 + 1.465 \times 10^{-3} T$, kJ/(kg.K) where T is in K.

C) Solve the following

6

- a. Calculate the standard heat of formation of n-propanol (C_3H_7OH) liquid using the following data: **U, A**
Standard heat of formation of $CO_{2(g)} = -393.51$ kJ/mol
Standard heat of formation of $H_2O_{(l)} = -285.83$ kJ/mol
Standard heat of combustion of n-propanol liquid = -2028.19 kJ/mol
- b. Calculate the heat that must be added to 3 kmol air to heat it from 298 K to 473 K using the mean molar heat capacity data for air given below: **U, A**
 m (between 473 K and 298 K) for air = 29.3955 kJ/kmol.K

Q. 5 Solve Any Two of the following.

12

- A) Give overall material balance for mixing in stirred tank (Unsteady state operation). **R, U** **6**
- B) A tank initially contains 500 kg of an aqueous solution of 10% salt. How much water must be fed to the tank to reduce the salt content to 1%? The contents of the tank are thoroughly mixed and the mass in the tank is maintained at 500 kg by keeping the continuous feed and drain streams at equal rates. **U, A** **6**
- C) A tank contains 5 m^3 of a 20 mass% solution of sodium chloride. Water is added at the rate of $3 \text{ m}^3\text{h}^{-1}$ and the solution is withdrawn at the same volumetric rate. Determine the mass of salt remaining after one hour. Assume thorough mixing in the tank and neglect density changes due to concentration. The tank can hold only 5 m^3 of solution. How long will it take for the mass per cent of the solute to become 5? **U, A** **6**

***** End *****

PSYCHROMETRIC CHART
NORMAL TEMPERATURE
SI METRIC UNITS
 Barometric Pressure 101.325 kPa
 SEA LEVEL

R = Reference Point

